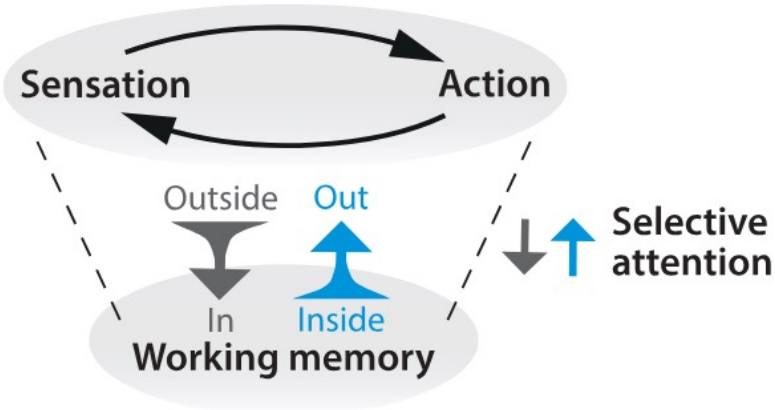
Modeling Goal-Directed Resource Allocation in Working Memory via Sparse Gaussian Process Approximation

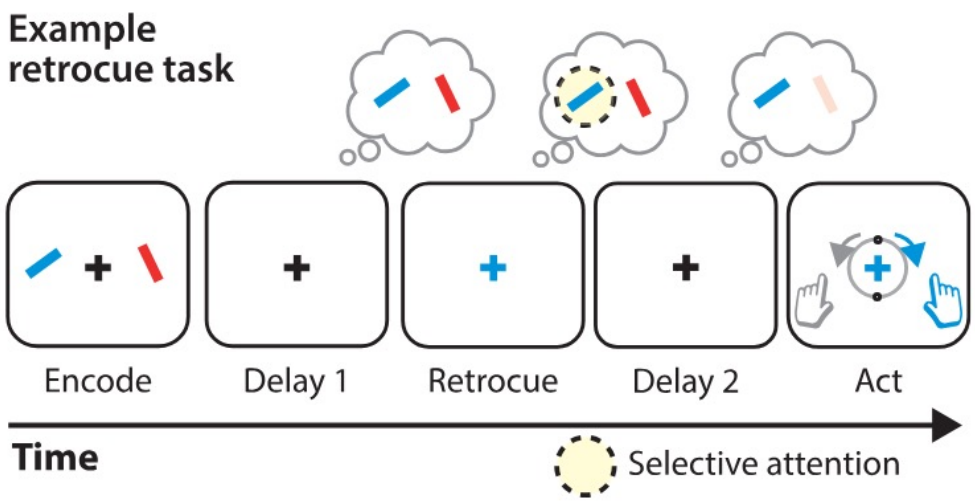
Dongyu Gong

Cog Neuro Joint Lab Meeting

Mar 6, 2026

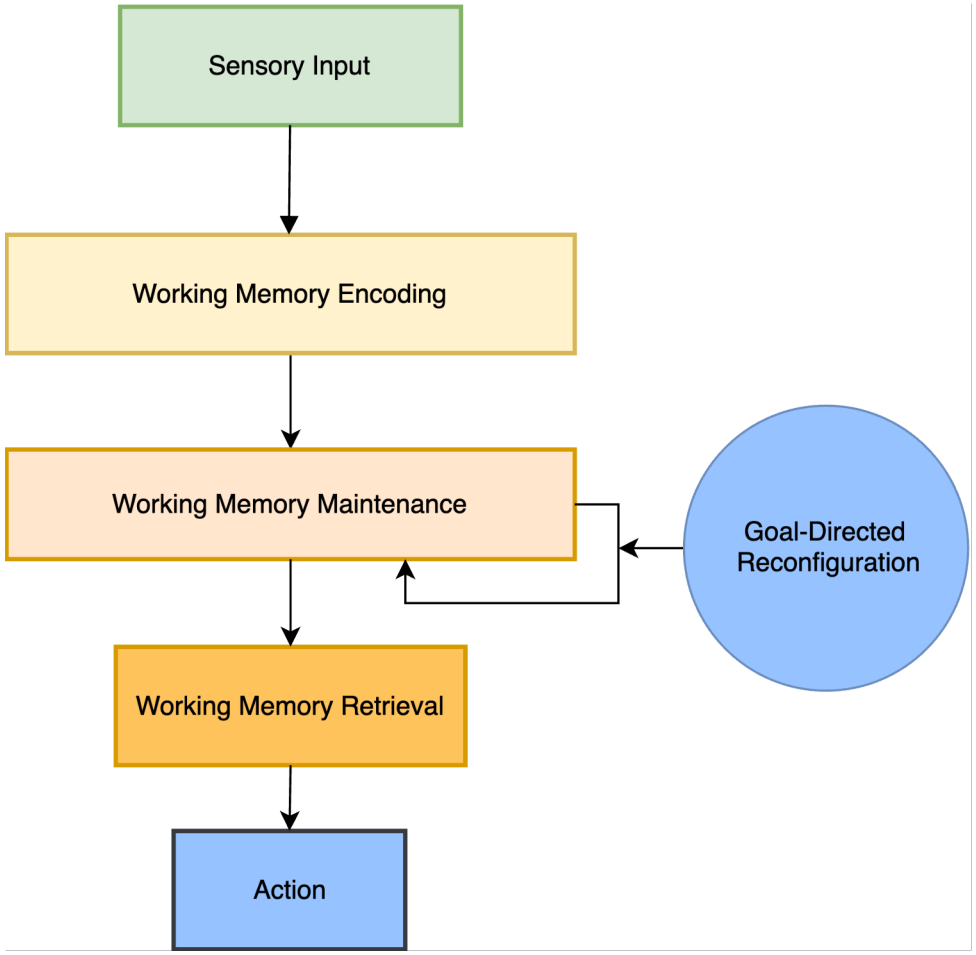
The Functional / Architectural View of WM



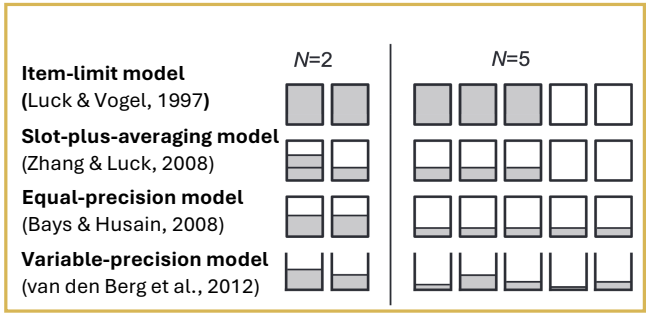


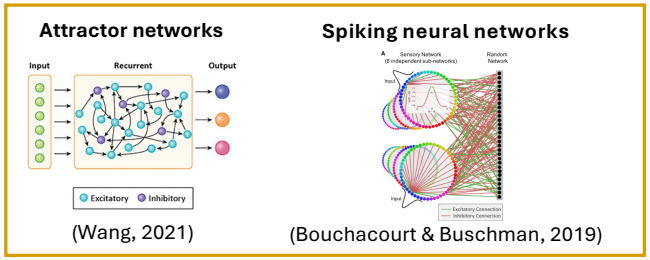
Griffin & Nobre, 2003 ; van Ede & Nobre, 2023

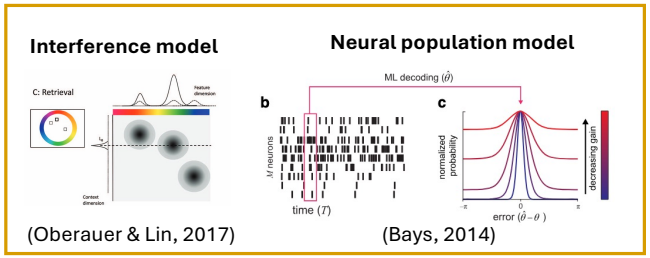
The “Life Cycle” of WM

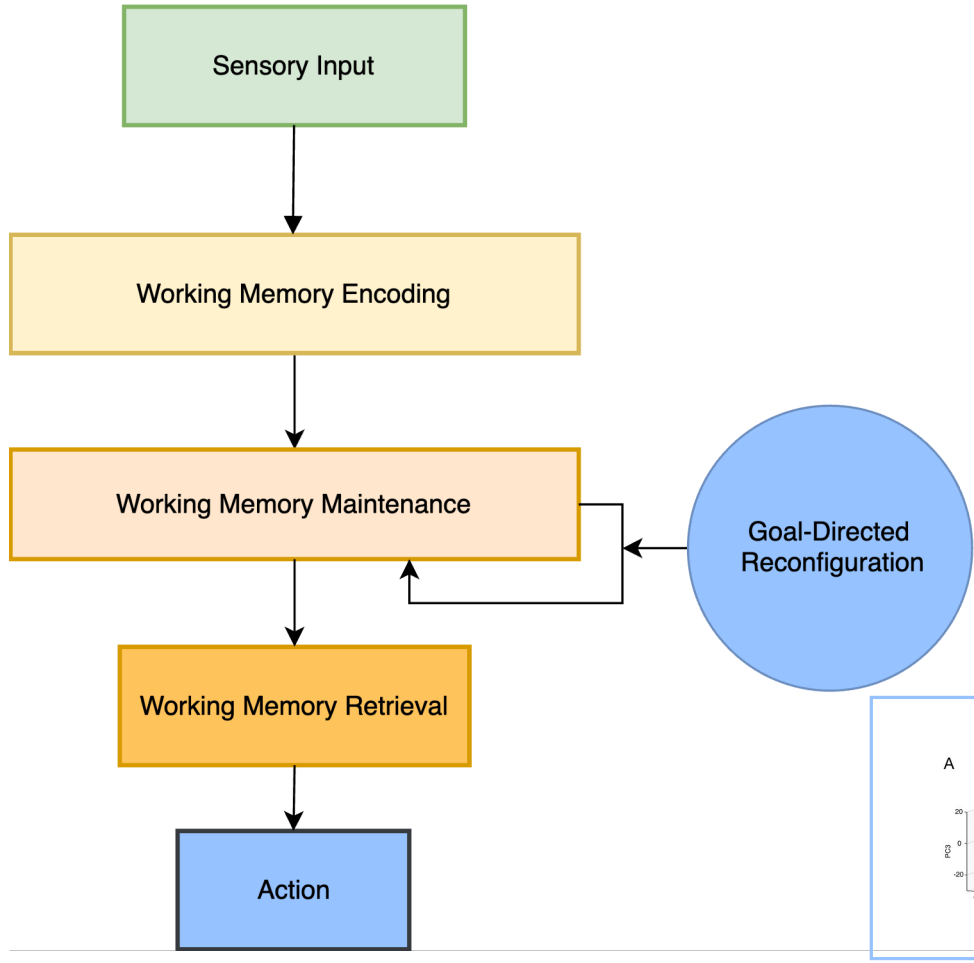


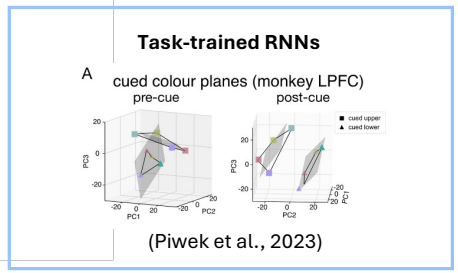
Piecemeal State of Modeling Work in WM



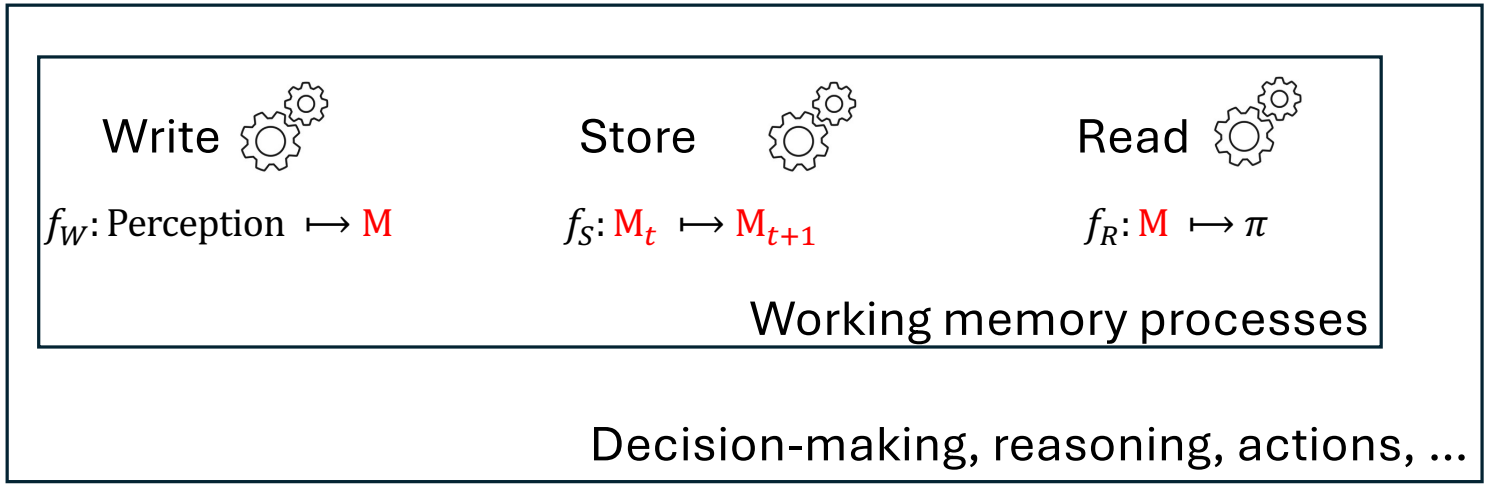






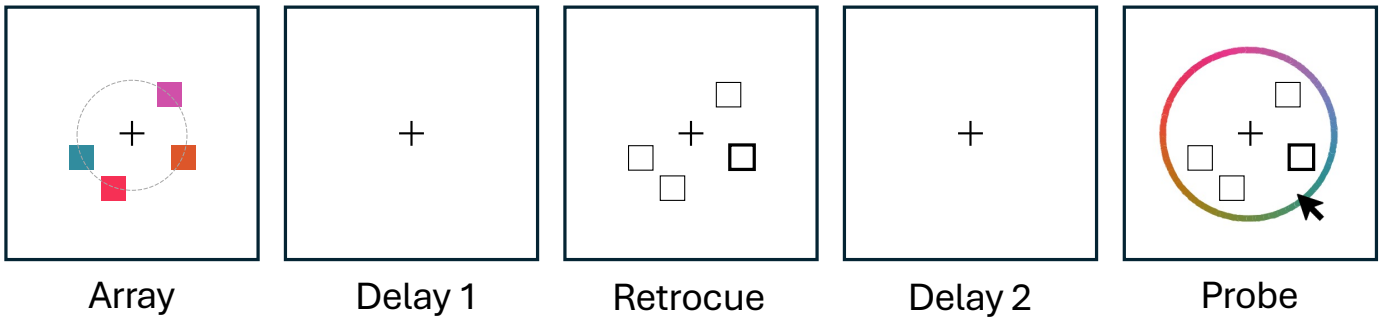


An Architectural / Algorithmic View of WM

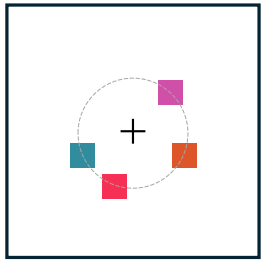


Example Task Domain

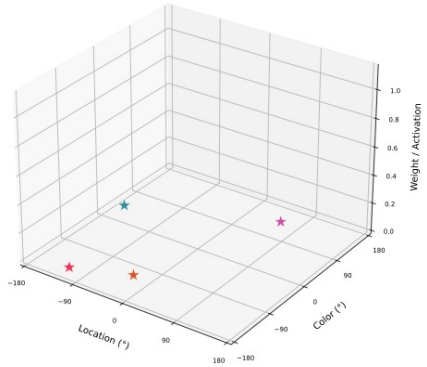
Continuous reproduction task with retrocues



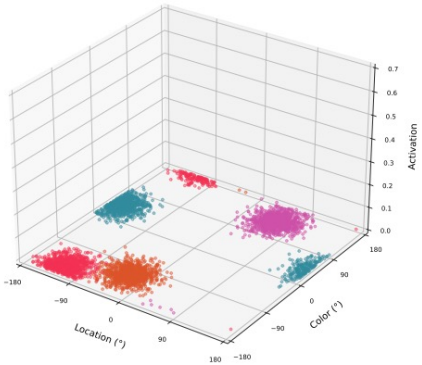
Write (WM Encoding)



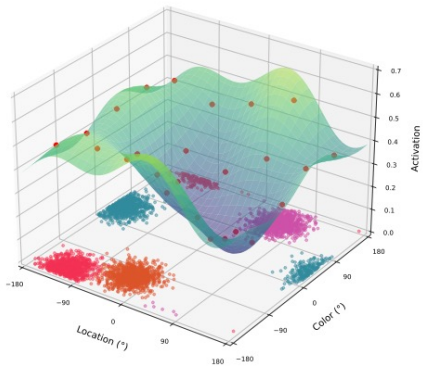
Array



True values



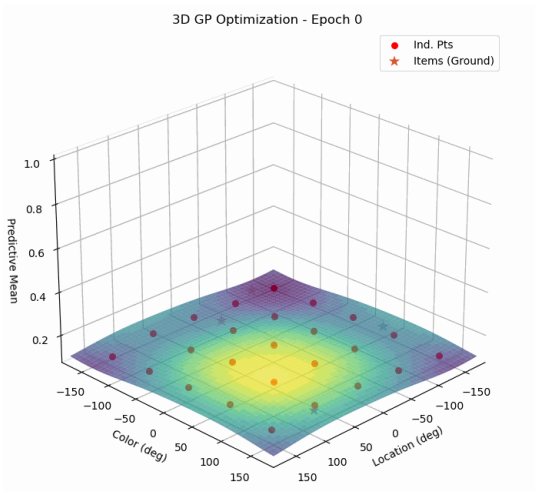
Sensory samples

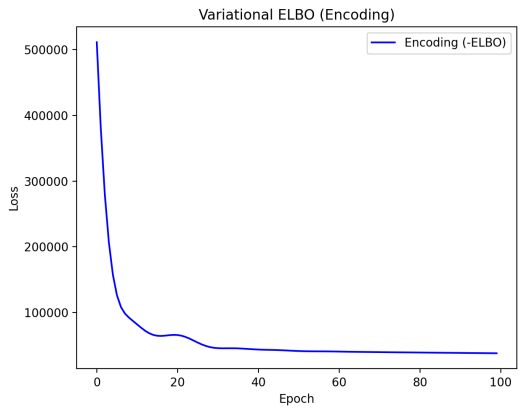


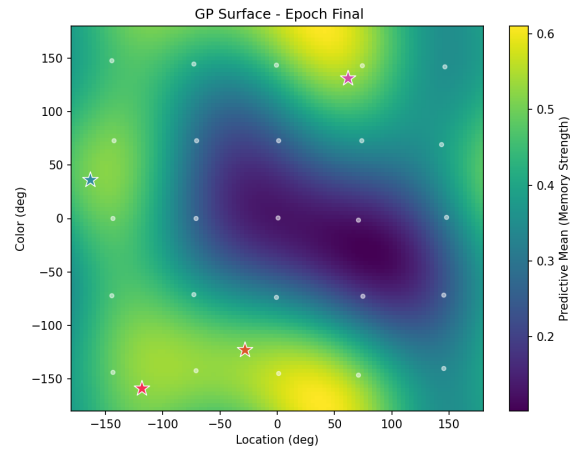
Write into WM

Write (WM Encoding)

- M : Targetable, shared resources that WM writes into, maintains, and retrieves
- Inducing Points in Gaussian Processes $f(x) \sim GP(\mathrm{m(x)}, \mathrm{k(x, x')})$







Objective function: $$\mathcal{L} = \sum_{i=1}^{n} \mathbb{E}_{q(f_i)} \left[\log p(y_i | f_i)\right] - \mathrm{KL}\left(q(\mathbf{u}) \| p(\mathbf{u})\right)$$

An Architectural / Algorithmic View of WM

Write

f_W: Perception $\longmapsto$ M

Store

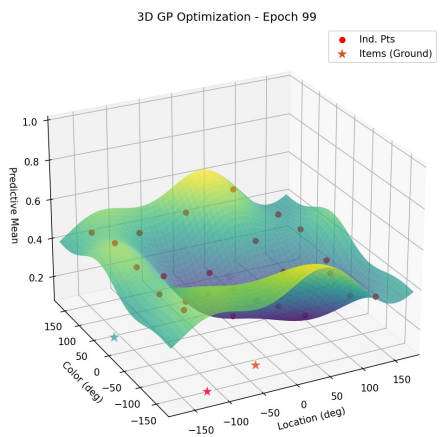
f_S: $\mathrm{M}_t \longmapsto \mathrm{M}_{t+1}$

Read

f_R: M $\longmapsto \pi$

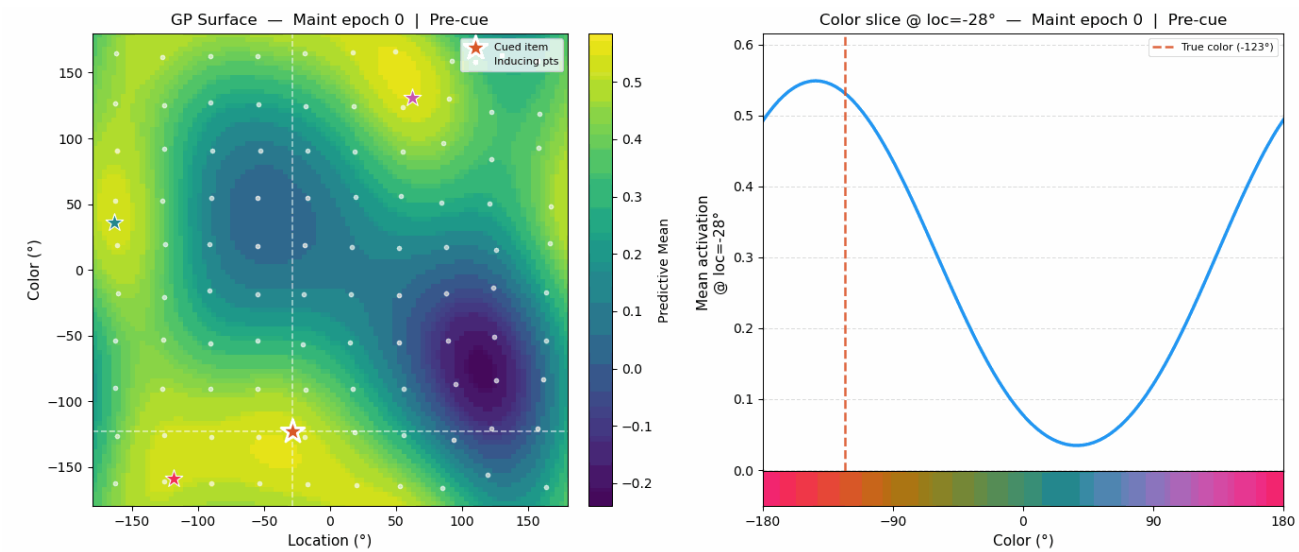
Working memory processes

Decision-making, reasoning, actions, ...



Store (WM Maintenance)

- Pre-cue: self-rehearsal (f_S: $\mathrm{M}_t \longmapsto \mathrm{M}_{t+1}$)



$$\mathcal{L}\text{pre-cue} = \left(\frac{0.5}{|X\text{maint}|} \sum_{\mathbf{x} \in X_{\text{maint}}} \mathbb{E}_{q(f(\mathbf{x}))}[\log p(y(\mathbf{x})|f(\mathbf{x}))] \right) - \beta \cdot \mathrm{KL}(q(\mathbf{u}) \| p(\mathbf{u}))$$

Adaptive Computation

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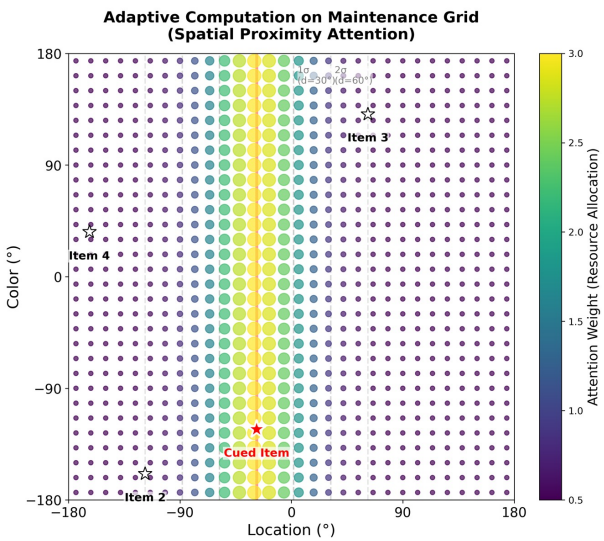
Adaptive Computation as a New Mechanism of Dynamic Human Attention

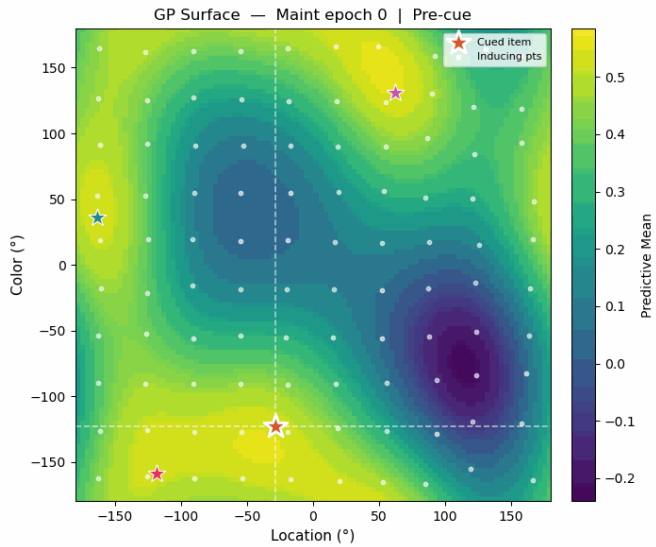
Mario Belledonne[1], Eivinas Butkus[2], Brian J. Scholl[1, 3], and Ilker Yildirim[1, 3, 4, 5]

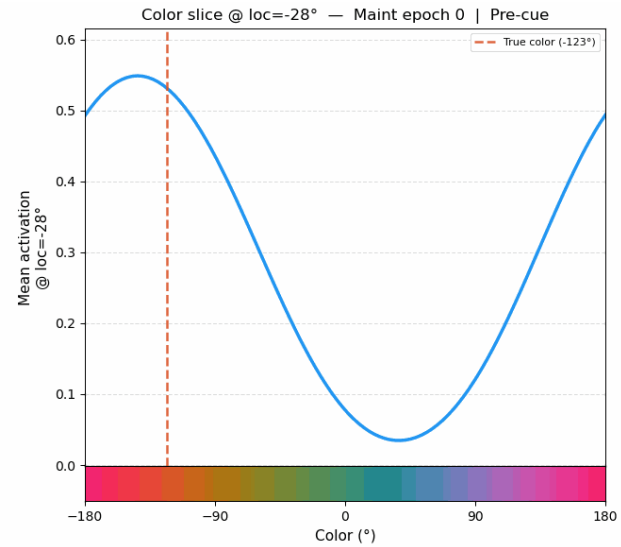
[1] Department of Psychology, Yale University
[2] Department of Psychology, Columbia University
[3] Wu Tsai Institute, Yale University
[4] Department of Statistics and Data Sciences, Yale University
[5] Foundations of Data Science Institute, Yale University

Selectively Store (Goal-driven Maintenance)

After retrocue: goal-conditioned rehearsal (f_S: $\mathrm{M}_t \longmapsto \mathrm{M}_{t+1}$)







$$w_{\text{post-cue}}(\mathbf{x}) = 0.5 + (\gamma - 0.5) \cdot \exp\left(-\frac{\mathrm{dist}(l_{\mathbf{x}}, l_{\text{cue}})^2}{2\sigma_{\text{attn}}^2}\right)$$

$$\mathcal{L}\text{post-cue} = \left(\frac{1}{|X\text{maint}|} \sum_{\mathbf{x} \in X_{\text{maint}}} w_{\text{post-cue}}(\mathbf{x}) \cdot \mathbb{E}_{q(f(\mathbf{x}))}[\log p(y(\mathbf{x})|f(\mathbf{x}))]\right) - \beta \cdot \mathrm{KL}(q(\mathbf{u}) \| p(\mathbf{u}))$$

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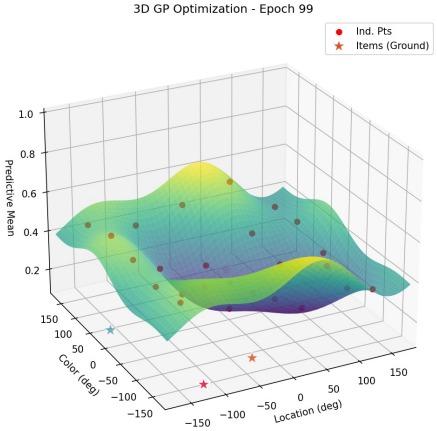
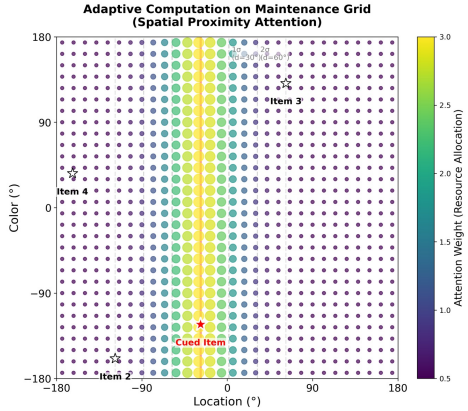
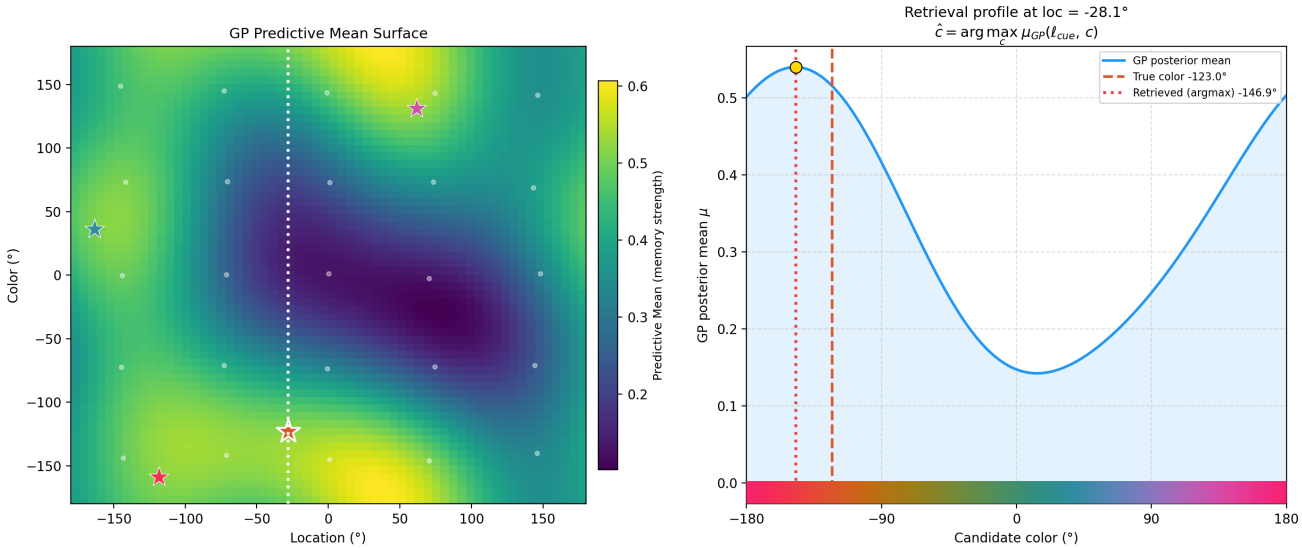




Read (WM Retrieval)

- Slice the GP based on the location probe and read out the color from the marginal predictive distribution f_R: $\mathrm{M} \longmapsto \pi$



$$\hat{c} = \arg\max_{c} \; \mu_{GP}(\ell_{cue}, \, c)$$

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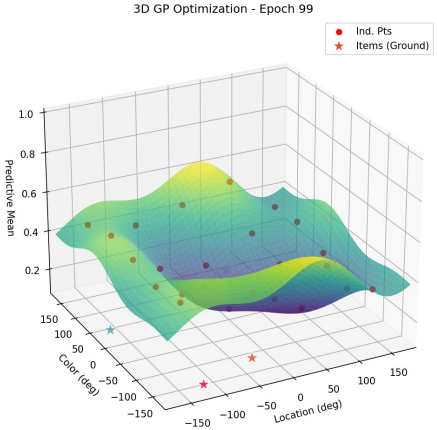
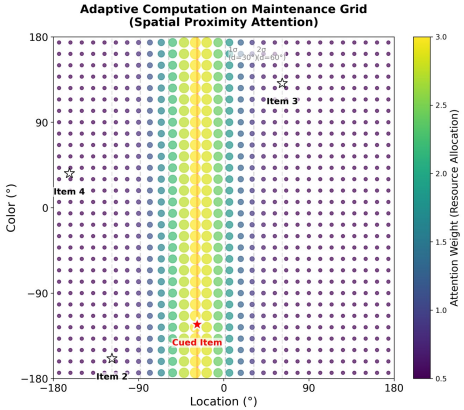
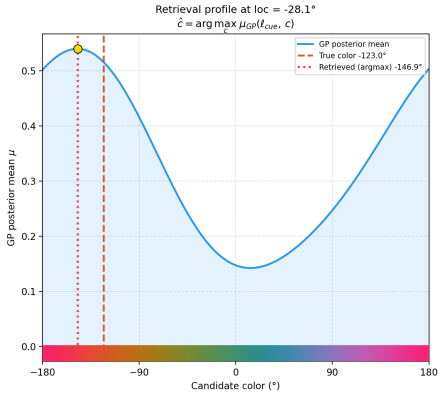
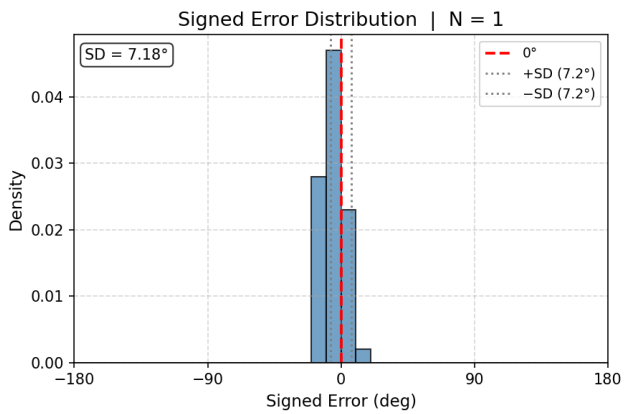
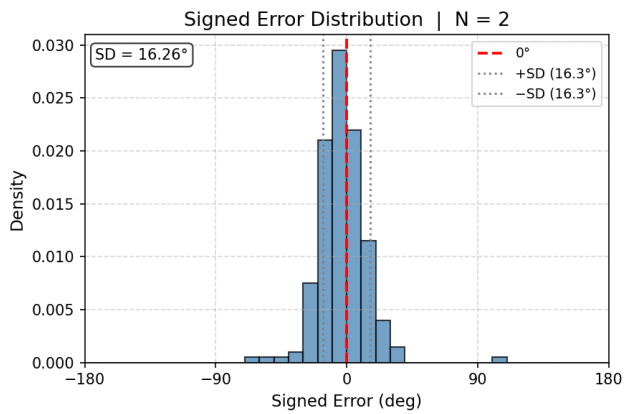


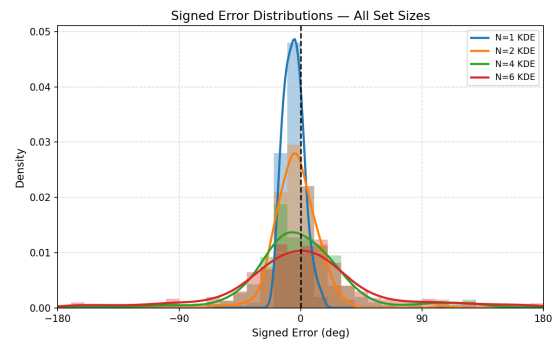


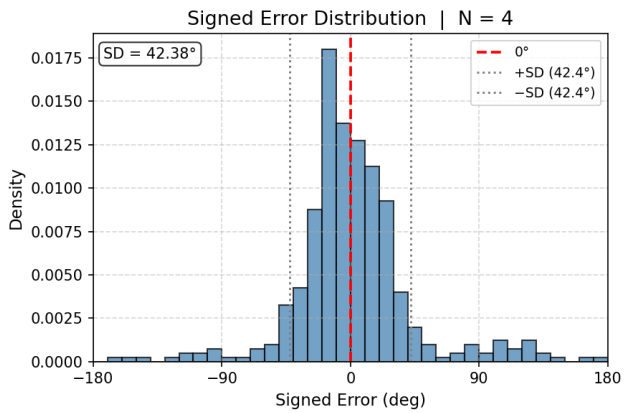


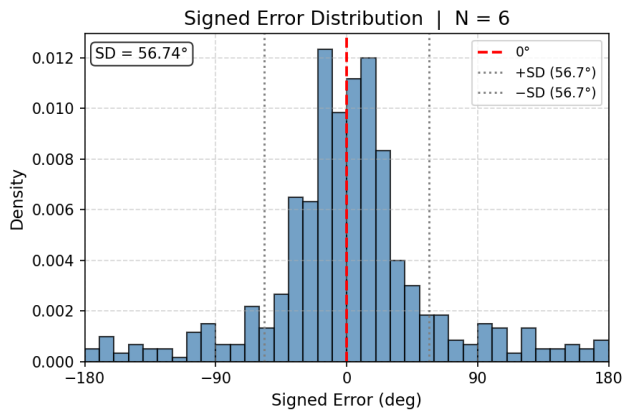
Set-size Effects

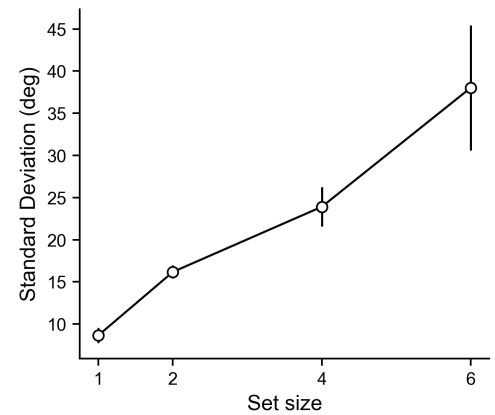




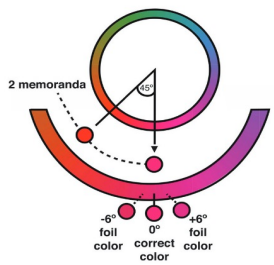




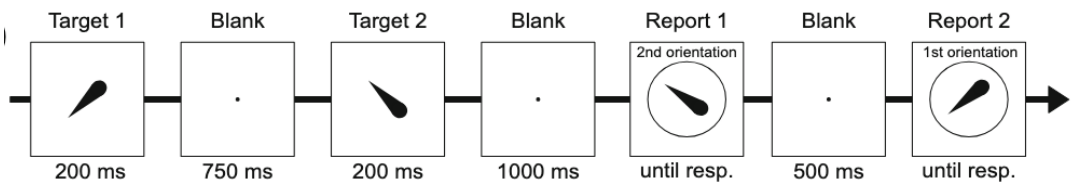




Attraction-repulsion Biases

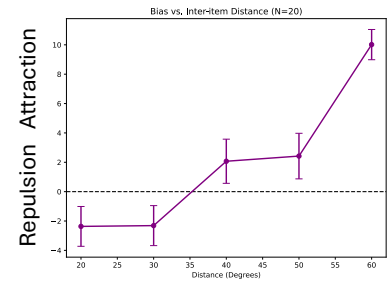


Chunharas et al. (2022)

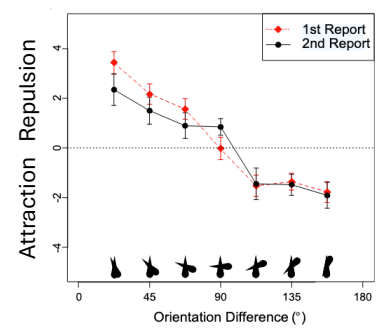


Bae & Luck (2017)

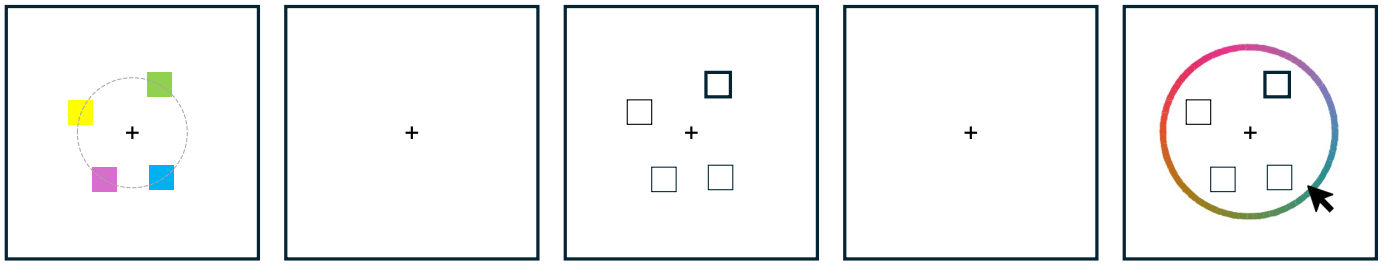
Model simulation



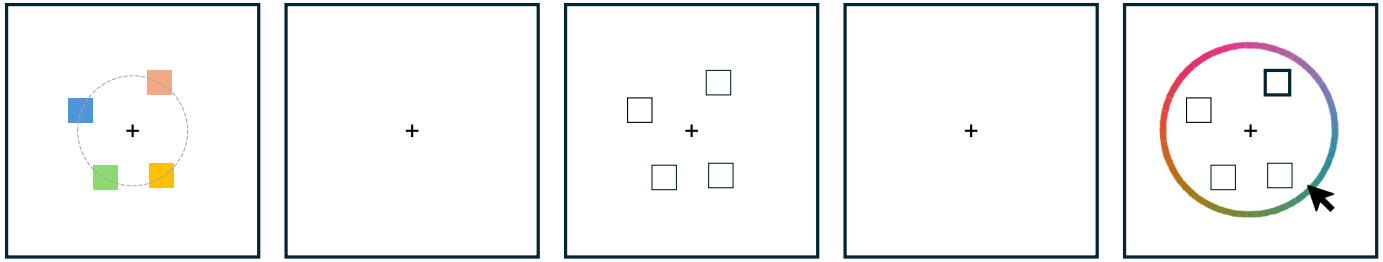
Human data



Retro-cue Benefits

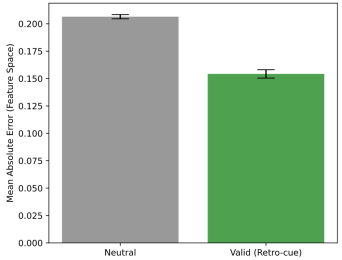


Valid cue

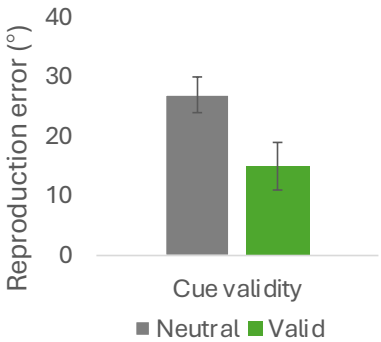


Neutral cue

Model simulation

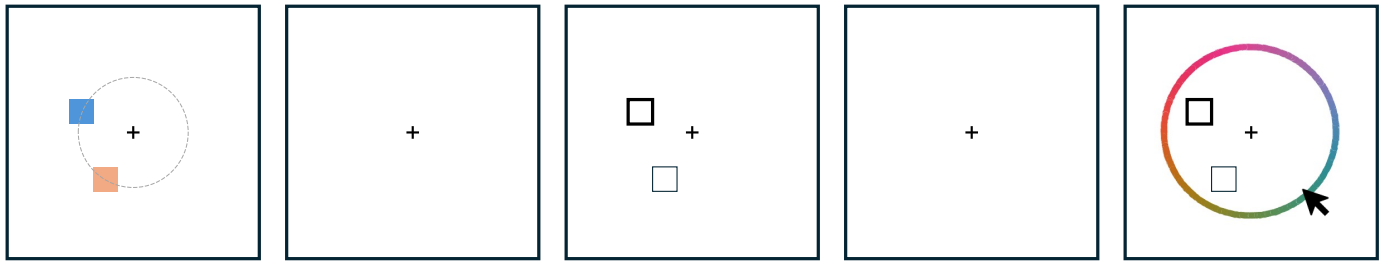


Human data



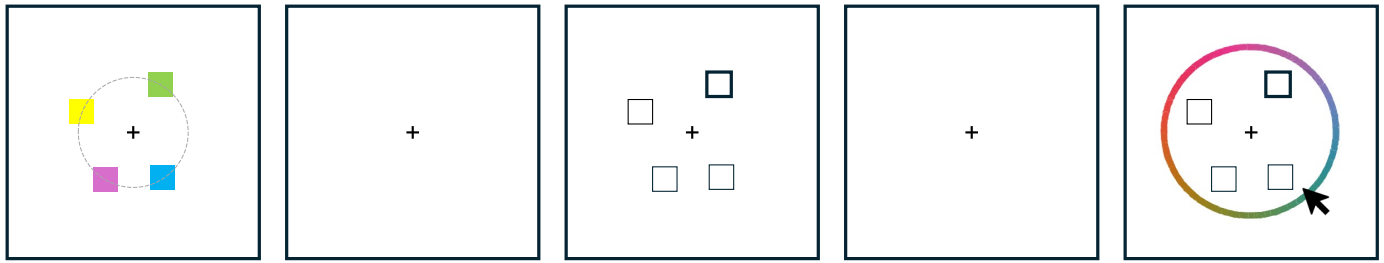
Prediction: Load-dependent Retro-cue Benefits

Set size = 2



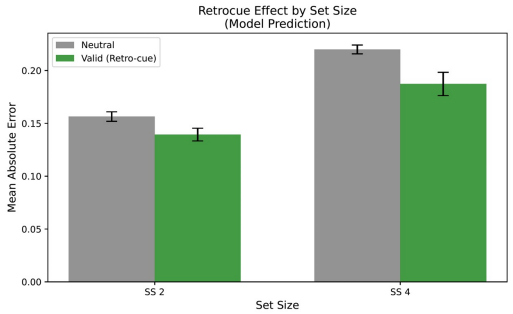
Retro-cue

Set size = 4

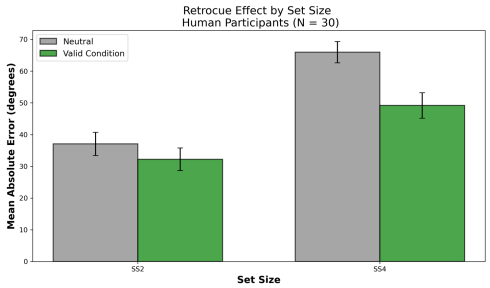


Retro-cue

Model prediction



Human data (N=30)



Conclusion

- In the current work, we provide an algorithmic account of **dynamic, goal-directed resource allocation in working memory** using sparse variational Gaussian processes.

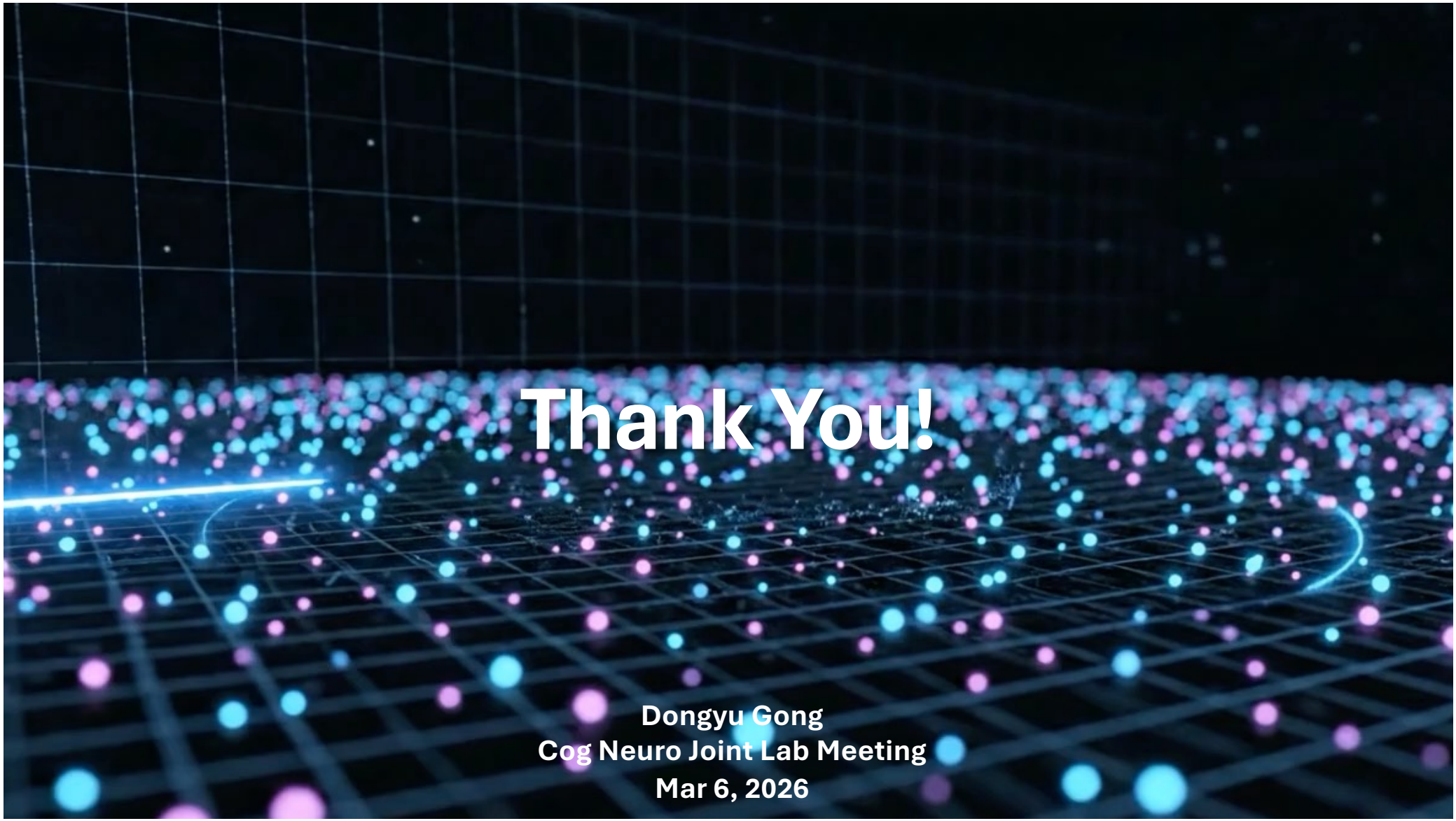
- We model the maintenance of visual information as an **online, goal-driven optimization process**, and the model has captured important phenomena in the WM literature and made novel predictions.

Acknowledgments

Mario Belledonne

Ilker Yildirim

Kia Nobre



Thank You!

Dongyu Gong

Cog Neuro Joint Lab Meeting

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